

Mixtures

A MIXTURE IS A COMBINATION OF SUBSTANCES THAT CAN BE SEPARATED BY PHYSICAL MEANS.

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Mixtures are classified as solutions, suspensions, or colloids based on particle size. The substances in a mixture are not chemically linked.

Uneven and even

Every mixture has at least two ingredients. The first ingredient is known as the continuous medium. Into this, the second ingredient, known as the dispersed phase, is mixed. In an even or homogeneous mixture, the particles of the dispersed phase are evenly distributed among the molecules of the continuous medium, so the concentrations of each ingredient are constant. In an uneven or heterogeneous mixture, the dispersed phase is concentrated in some places and not in others. Some substances, normally liquids, cannot be mixed together because their molecules repel each other—they are described as immiscible.

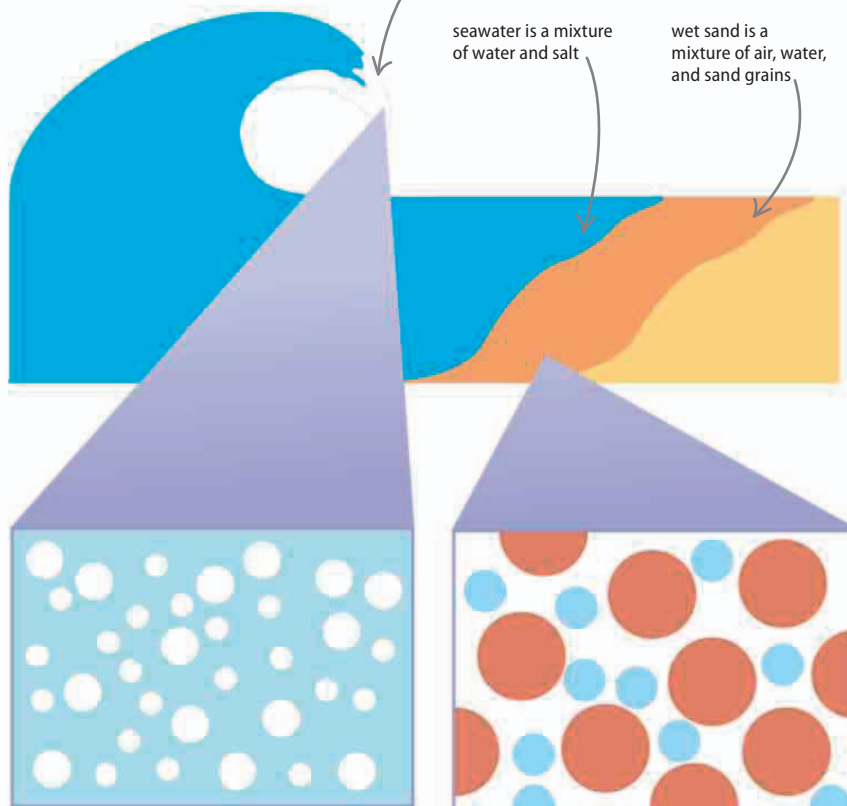
▽ Seawater

In seawater, the water is the continuous medium, while salts—chiefly sodium chloride—form the dispersed phase. The salt mixes so thoroughly that it dissolves and disappears from view.

foam is a mixture of air bubbles and water

seawater is a mixture of water and salt

wet sand is a mixture of air, water, and sand grains



△ Foam

The foam of a breaking wave is a heterogeneous mixture of air bubbles and water. The white appearance of the mixture is different from those of both the constituents.

△ Wet sand

The sand grains are much larger than the water molecules around them so they remain distinct and are visible when viewed close up. As the mixture dries, the water is replaced by air.

REAL WORLD

Lava lamp

A lava lamp makes use of two immiscible liquids. The clear liquid is a mineral oil, while the colored “lava” is a wax. When the lamp is turned on, light heats the wax, reducing its density so it begins to rise up into the oil. The wax does not mix, however, and the colored bubbles rise and fall.

